Project Overview:

This project will help students build a full-fledged interactive dashboard in Power BI that analyzes how students' social media usage correlates with sleep patterns, academic performance, mental health, and relationship dynamics.

Stakeholders:

- **Academic Advisors** – interested in academic performance and social media usage
- **Mental Health Counselors** – concerned with sleep, mental health, and addiction scores
- **Institutional Researchers** – want country- and gender-level insights
- **Students & Counselors** – view specific profiles for drill-through insights

Project Structure:

Phase 1: Data Loading, Modeling, and Cleaning

Task 1: Load the Dataset

- Load both sheets from Excel: Student Details and Platform Details
- Check data types and consistency

Task 2: Create Relationships

- Join both tables using Student_ID
- Verify a **1:1 relationship**
- Ensure both tables are in **Star Schema** (fact = behavior metrics, dimension = demographics)

Task 3: Add a Custom Date Table (for mock date-based analysis)

- Create a simple date table using CALENDAR() for future use

```
DateTable = CALENDAR(DATE(2023, 1, 1), DATE(2023, 12, 31))
```

- Mark as a date table

Phase 2: Data Preparation and Transformation

Task 4: Create Calculated Columns

- Health Band = IF(Mental_Health_Score >= 7, "Good", IF(Mental_Health_Score >= 4, "Average", "Poor"))
- Conflict Level = SWITCH(TRUE(), Conflicts_Over_Social_Media > 5, "High", Conflicts_Over_Social_Media > 2, "Medium", "Low")

Task 5: Create Measures

- Avg Sleep Hours = AVERAGE(Student_Details[Sleep_Hours_Per_Night])
- Avg Usage Hours = AVERAGE(Platform_Details[Avg_Daily_Usage_Hours])
- % Affected Academically = DIVIDE(CALCULATE(COUNTROWS(Platform_Details), Platform_Details[Affects_Academic_Performance] = "Yes"), COUNTROWS(Platform_Details))
- Addicted Student Count = CALCULATE(COUNTROWS(Student_Details), Student_Details[Addicted_Score] > 7)

Phase 3: Time Intelligence (Simulation)

Task 6: Use Date Table (Optional Time-Based Extension)

- Create:
  - Monthly Trend = CALCULATE([Avg Usage Hours], DATESYTD(DateTable[Date]))

Phase 4: Report Pages and Visuals

Page 1: Executive Overview

- Card visuals:
  - Total Students
  - Average Usage
  - % Affected Academically
  - Avg Sleep Hours

- Bar chart: Addiction Score by Academic Level
- Line chart: Avg Usage Hours by Age
- Pie chart: Gender Distribution

Page 2: Mental Health & Lifestyle

- Matrix: Country vs Health Band
- Scatter chart: Addicted Score vs Mental Health Score
- Line chart: Sleep Hours vs Age

Page 3: Academic Impact

- Bar chart: Avg Daily Usage vs Academic Level
- Column chart: Academic Performance Impact by Platform

Page 4: Relationships and Conflicts

- Stacked bar: Conflict Level by Relationship Status
- Table: Student wise Addiction Score
- Pie: Relationship Status Distribution

Page 5: Interactive Story View

- Bookmark View 1: Social Media by Gender
    - Bar Chart: Avg_Daily_Usage_Hours by Gender
    - Donut Chart: Most_Used_Platform by Gender
- Bookmark View 2: Social Media by Academic Level
    - Bar Chart: Avg_Daily_Usage_Hours by Academic_Level
    - Pie Chart: Most_Used_Platform by Academic_Level
- Go to View > Bookmarks Pane and Selection Pane
- For Bookmark 1 (Gender View):
    - Hide the Academic Level visuals
    - Show the Gender visuals
    - Click "Add", name it Gender View

- For Bookmark 2 (Academic Level View):
  - Do the opposite: hide Gender visuals, show Academic Level visuals
  - Click "Add", name it Academic Level View
- Create two buttons to show Gender View and Academic Level View

Page 6: Drill-through Page: Student Profile

- Create a new page for Drill-through to Student_ID
- Drill-through on Student_ID
- Show:
  - Student Demographics
  - Sleep Hours
  - Usage Hours
  - Addicted Score
  - Most Used Platform
  - Conflicts
  - Mental Health Rating

Phase 5: Interactivity and Storytelling

Task 7: Slicers & Filters

- Add slicers for:
  - Mental Health & Lifestyle - Gender
  - Relationships & Conflicts – Country, Academic Level
  - Academic Impact - Age Range (using numeric slider)